INVESTMENT PLATFORM WITH AI-POWERED STOCK MARKET INSIGHTS FOR COLOMBO STOCK EXCHANGE

(OPEN PRICE PREDICTION COMPONENT)

25-26J-445

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DECLARATION

I declare that this is my own work and this dissertation does not incorporate without acknowledgement any material previously submitted for a Degree or Diploma in any other University or institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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ABSTRACT

Short-term stock price forecasting in emerging markets is a complex challenge due to market non-linearity, limited liquidity, and high volatility. This dissertation presents an ensemble-based machine learning framework designed to predict five-trading-day-ahead (T+5) open prices for all 264 companies listed on the Colombo Stock Exchange (CSE). Daily OHLCV data spanning 2019 to 2024 were collected from official CSE sources, preprocessed to address missing values and non-stationarity, and transformed using log returns to stabilise variance. A total of 28 engineered features — comprising momentum indicators, moving averages, volume signals, volatility measures, and company-level statistics — were derived from raw historical data to capture diverse market dynamics.

An equally weighted ensemble of three gradient boosting models — XGBoost, LightGBM, and CatBoost — was proposed and evaluated against six alternative models including statistical baselines and individual gradient boosting variants. The ensemble achieved a directional accuracy of 61.44% on the out-of-sample test set (July to December 2024), representing an improvement of 11.44 percentage points over random guessing and outperforming all individual models. Price-level predictions yielded an R^2 of 0.9829 and a median absolute percentage error of 2.76% across 22,996 predictions. Cross-sectional analysis across all 264 CSE companies confirmed model robustness, with a median company-level directional accuracy of 57.37% and near-zero prediction bias of +0.021%.

To enhance transparency and institutional trust, post-hoc Explainable AI (XAI) techniques using SHAP values were integrated to identify key predictive features per prediction. The forecasting service was deployed as a FastAPI microservice within a modular, containerised microservices architecture, with an automated daily data pipeline implemented via GitHub Actions for continuous model inference. The resulting system provides a scalable, interpretable, and operationally reliable decision-support tool for retail investors and analysts operating within the CSE ecosystem.

Keywords - open price prediction, ensemble learning, gradient boosting, Colombo Stock Exchange, XGBoost, LightGBM, CatBoost, SHAP, explainable AI, financial forecasting, microservices.

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I also extend my sincere appreciation to the Department of Computer Science and Software Engineering at SLIIT for providing access to computing resources, research infrastructure, and a stimulating academic environment that made this research possible.

Finally, I acknowledge the open-source community behind XGBoost, LightGBM, CatBoost, FastAPI, and SHAP, whose freely available tools formed the technical backbone of this system.

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LIST OF ABBREVIATIONS

Abbreviation	Description
ADF	Augmented Dickey-Fuller
API	Application Programming Interface
ARIMA	Autoregressive Integrated Moving Average
CSE	Colombo Stock Exchange
CI/CD	Continuous Integration / Continuous Deployment
EMA	Exponential Moving Average
GARCH	Generalised Autoregressive Conditional Heteroskedasticity
HTTP	Hypertext Transfer Protocol
JWT	JSON Web Token
LKR	Sri Lankan Rupee
MACD	Moving Average Convergence Divergence
MAE	Mean Absolute Error
ML	Machine Learning
OHLCV	Open, High, Low, Close, Volume
REST	Representational State Transfer
RSI	Relative Strength Index
SHAP	SHapley Additive exPlanations
SMA	Simple Moving Average
T+5	Five Trading Days Ahead
XAI	Explainable Artificial Intelligence

1 INTRODUCTION

The global financial landscape has increasingly embraced technology-driven decision support, with machine learning (ML) becoming a cornerstone of modern quantitative analysis. In emerging markets such as Sri Lanka, retail investors and institutional analysts face compounded challenges: low liquidity, higher volatility, and limited access to sophisticated analytical tools. The Colombo Stock Exchange (CSE), while growing in participation and trading volume, remains underserved by advanced forecasting infrastructure compared to developed capital markets.

This dissertation describes the design, development, and evaluation of the Open Price Prediction component within a broader AI-powered investment platform built for the CSE. The component addresses the need for reliable, short-term price forecasting by predicting the opening price of a stock five trading days (T+5) into the future. The T+5 horizon aligns with practical investment windows of retail investors who make weekly portfolio adjustment decisions.

The broader investment platform — developed collaboratively by a four-member research team — integrates financial health analysis, risk prediction, dividend forecasting, and sentiment analysis alongside the open price prediction module. Each component is deployed as an independent microservice, communicating through a NestJS API Gateway. This dissertation focuses exclusively on the individual contribution of the open price prediction component.

1.1 Background

Financial time-series forecasting has a long history in both academic research and industry practice. Traditional econometric models such as ARIMA and GARCH provided foundational approaches to modelling autocorrelation and conditional volatility in price data. However, the non-linear and non-stationary nature of modern equity markets has significantly limited the practical effectiveness of these models,

particularly during periods of structural breaks, sudden liquidity shocks, or market-wide sentiment shifts [1].

The emergence of machine learning methods — particularly tree-based ensemble algorithms such as Random Forest, XGBoost, LightGBM, and CatBoost — has introduced a paradigm shift in financial prediction tasks. These models excel at capturing complex, non-linear feature interactions without requiring the distributional assumptions imposed by traditional statistical methods [2]. In emerging market contexts, where data quality and market efficiency differ substantially from developed markets, such flexibility is particularly valuable.

The CSE, established in 1896 and now comprising over 270 listed companies across 20 business sectors, presents a unique testbed for applying ML-based forecasting. Prior work on CSE-specific prediction has largely employed univariate time-series methods or simple regression approaches [4]. The integration of multi-feature technical analysis with gradient boosting represents a more comprehensive and practically deployable approach for this market context.

1.2 Literature Review

For a long time, the field of financial forecasting was dominated by statistical time-series models like GARCH and ARIMA, which were primarily used to interpret volatility and linear price movements. However, the erratic nature and inherent non-linearity of modern markets often cause these traditional tools to fall short, particularly during high-volatility periods when short-term accuracy is most critical [1]. This limitation has sparked a significant shift in recent scholarship toward ensemble-based machine learning. Specifically, gradient boosting architectures — most notably XGBoost, LightGBM, and CatBoost — have emerged as the new standard [2]. By combining multiple weak learners through sophisticated bagging and boosting techniques, these models effectively refine predictive accuracy while simultaneously mitigating the high variance often found in single-model approaches [3].

When looking at emerging markets like the Colombo Stock Exchange (CSE), contemporary research highlights how much predictive power can be gained by integrating technical indicators — such as momentum and volume signals — to interpret sudden price shifts [4]. CatBoost, in particular, has distinguished itself through its ordered boosting logic; this mechanism is vital for neutralising gradient bias and managing categorical data without the need for heavy-handed preprocessing [5]. However, the modern academic consensus is that raw accuracy is no longer sufficient. The rise of Explainable Artificial Intelligence (XAI), utilising tools like SHAP and LIME, is now viewed as a prerequisite for building the institutional trust and regulatory transparency necessary for real-world deployment [6].

Table 1: Comparison of State-of-the-Art Stock Price Prediction Methods

Method	Approach	Market Focus	XAI Support	Key Limitation
ARIMA/GARCH	Statistical time-series	Developed markets	No	Assumes linearity; poor in volatile periods
Random Forest	Bagged decision trees	General	Partial (feature importance)	Lower directional accuracy than boosting ensembles
XGBoost	Gradient boosting	General	Partial (SHAP)	Requires careful hyperparameter tuning
LightGBM	Leaf-wise gradient boosting	General	Partial (SHAP)	Can overfit on small datasets
CatBoost	Ordered gradient boosting	Categorical-heavy data	Partial (SHAP)	Slower training on large datasets
Proposed Ensemble	Equal-weight XGBoost + LightGBM + CatBoost	CSE (Emerging market)	Yes (SHAP per prediction)	Dependent on historical data availability

Despite these strides, there is a noticeable void in research concerning automated, interpretable ensemble pipelines specifically calibrated for the unique dynamics of the CSE. This study addresses this gap by introducing a trend-aware, robust framework for five-day-ahead open price forecasting, combining ensemble precision with real-time operational reliability.

1.3 Research Gap

A review of the existing literature reveals several important gaps that this research aims to address:

- The majority of existing CSE-focused studies rely on ARIMA, GARCH, or simple regression techniques that do not exploit the predictive power of rich multi-feature technical analysis.
- Ensemble gradient boosting methods that are well established in developed market contexts have not been systematically applied to the full universe of CSE-listed companies.
- Most prior work generates point predictions without providing explanation or uncertainty quantification, limiting their practical utility for regulatory and retail investment decision support.
- There is an absence of deployed, production-ready forecasting microservices for the CSE that support continuous daily inference with automated data pipelines.
- No existing system combines T+5 open price forecasting with SHAP-based per-prediction explanation for the CSE context.

This research directly addresses these gaps by proposing an equal-weight ensemble of XGBoost, LightGBM, and CatBoost, trained and evaluated on six years of CSE data (2019–2024) across all 264 listed companies, combined with SHAP-based explainability and a fully automated deployment pipeline.

1.4 Research Problem

Retail and institutional investors in the CSE lack access to reliable, automated, and interpretable short-term price forecasting tools tailored to the unique characteristics of this emerging market. Existing approaches either rely on outdated statistical methods that assume linearity, or employ black-box ML models that offer no insight into the basis of their predictions, eroding user trust and limiting regulatory acceptance.

The specific research problem is: How can a gradient boosting ensemble model, trained on CSE-specific historical OHLCV data and engineered technical features, achieve superior directional accuracy for T+5 open price forecasting, while simultaneously providing SHAP-based prediction explanations and operating within a scalable, production-ready microservices architecture?

1.5 Research Objectives

The primary objective of this research is to design, develop, and evaluate an ensemble-based machine learning service for predicting five-trading-day-ahead open prices of all CSE-listed equities, incorporating explainable AI and production deployment capabilities.

Specific objectives include:

1. To collect, clean, and preprocess six years of daily OHLCV data for all 264 companies listed on the Colombo Stock Exchange (2019–2024).
2. To engineer 28 technical and company-level features from raw OHLCV data to capture momentum, volatility, volume, and seasonal dynamics.
3. To train and optimise XGBoost, LightGBM, and CatBoost models using time-series-aware walk-forward cross-validation and evaluate their individual predictive performance.
4. To construct an equally weighted ensemble of the three gradient boosting models and demonstrate its superior directional accuracy over individual models and statistical baselines.
5. To integrate SHAP-based post-hoc explanation into the prediction pipeline to provide per-prediction feature attribution.
6. To deploy the forecasting service as a FastAPI microservice with an automated daily data and inference pipeline, and integrate it with the NestJS API Gateway of the broader investment platform.
7. To evaluate system performance on an out-of-sample test set and validate robustness through cross-sectional analysis across all 264 CSE companies.

2 METHODOLOGY

2.1 System Overview

The open price prediction component forms one of five specialised analytical modules within a broader AI-powered investment platform for the CSE. The overarching platform is built as a microservices architecture, with a NestJS API Gateway (running on port 3400) serving as the single entry point for all client requests. The API Gateway routes incoming requests to the appropriate backend service using a combination of HTTP proxy calls and gRPC-based inter-service communication.

The open price prediction service — the subject of this dissertation — is implemented as a FastAPI microservice running on port 5002 within a Docker container. It operates independently of other platform services and exposes a REST API consumed by the API Gateway, which in turn serves the Next.js frontend application. This modular separation ensures that model updates, data pipeline changes, or infrastructure scaling within the prediction service do not affect the stability of other platform components.

2.1.1 Components of the System

The overall investment platform integrates five core analytical modules, each developed as an independent microservice:

- Financial Health Service (NestJS, HTTP :3001): Computes Altman Z-scores and financial health metrics for listed companies based on submitted financial ratios.
- Risk Prediction Service (Flask ML, HTTP :5001): Employs a Support Vector Regression model to predict company-level financial risk scores.
- Open Price Prediction Service (FastAPI ML, HTTP :5002): Predicts T+5 open prices using an ensemble of gradient boosting models. This is the component described in this dissertation.
- Dividend Prediction Service (FastAPI ML, HTTP :8001): Forecasts future dividend payments for CSE-listed companies.

- Sentiment Analysis Service (FastAPI ML, HTTP :8002): Performs two-stage sentiment analysis on financial news using OpenAI integration.

All services are orchestrated through the NestJS API Gateway, which handles JWT authentication, request routing, and error translation between gRPC status codes and HTTP responses. Infrastructure services (Apache Kafka on port 29092 and Zookeeper on port 22181) support asynchronous email notification workflows.

[NOTE TO AUTHOR: Insert Figure 1 — Screenshot of the overall system architecture diagram here. A screenshot from the system's architecture documentation or a diagram showing all microservices and their ports would be appropriate.]

2.1.2 Technological Stack

The open price prediction component employs a carefully selected set of technologies optimised for ML model serving, feature engineering, and integration with the broader platform infrastructure:

Machine Learning and Data Processing:

- Python 3.11: Core language for all ML development, data processing, and inference scripting.
- XGBoost, LightGBM, CatBoost: Gradient boosting frameworks used for model training. Models are serialised using scikit-learn's pickle protocol and stored as .pkl files.
- Pandas and NumPy: Used for data manipulation, feature engineering, and numerical computation.
- SHAP (SHapley Additive exPlanations): Post-hoc model explainability using TreeExplainer for all three sub-models.
- scikit-learn: Provides LabelEncoder for company ID encoding, cross-validation utilities, and evaluation metrics.

Service Framework and Deployment:

- FastAPI: Asynchronous Python web framework for building the prediction REST API with automatic Swagger documentation.
- Docker: Containerisation of the ML service, with libgomp1 included in the runtime stage for LightGBM OpenMP support.
- GitHub Actions: Automated CI/CD pipeline for daily data updates and model inference.

Integration:

- NestJS (API Gateway): HTTP proxy layer that routes frontend requests to the FastAPI service, providing a 120-second timeout for prediction calls and error mapping.
- Next.js (Frontend): Renders prediction results, SHAP explanations, and historical OHLCV charts in the user interface.
- Zustand: Client-side state management for prediction results in the React-based frontend.
- Recharts: Frontend charting library used to visualise historical prices, prediction comparisons, and SHAP driver bars.

2.2 Feasibility Study

2.2.1 Initial Planning and Scope Definition

The initial planning phase involved defining the scope of the open price prediction component in relation to the broader investment platform. Requirements were defined in consultation with the project team and academic supervisors, focusing on the practical needs of CSE retail investors and analysts. The core requirement was a system capable of providing weekly (T+5) open price forecasts for all actively listed CSE companies, with associated uncertainty indication and prediction explanation.

The scope was defined to include: data collection and preprocessing (6 years of CSE OHLCV data), feature engineering (28 technical and company-level features), model training and ensemble construction, SHAP-based explainability, API deployment, and frontend integration. Out-of-scope for this component were fundamental analysis,

news sentiment integration (handled by a separate service), and real-time tick data processing.

A phased development schedule was adopted aligned with the academic calendar: data collection and preprocessing (Phase 1), feature engineering and model development (Phase 2), evaluation and XAI integration (Phase 3), and deployment and frontend integration (Phase 4).

2.2.2 Risk Assessment and Mitigation

A formal risk assessment was conducted to identify potential development risks and define mitigation strategies. Table 2 summarises the identified risks for the open price prediction component.

Table 2: Risk Analysis for the Open Price Prediction Component

Risk ID	Description	Impact	Likelihood	Mitigation Strategy
R1	Insufficient historical OHLCV data for some CSE companies	High	Medium	Use the full 2019–2024 dataset; exclude companies with fewer than 30 trading rows
R2	Model overfitting on training data	High	Medium	Apply walk-forward cross-validation; use regularisation (L1/L2) in all models
R3	Ensemble model feature dimension mismatch	High	Low	Validate feature count (28) at startup; use _pad as 28th column for all models
R4	SHAP explainer slow on first initialisation	Medium	High	Lazy-initialise SHAP TreeExplainer on first explain=true request and cache
R5	FastAPI service unavailable during Docker restart	Medium	Low	Implement health-check endpoint; API Gateway returns 503 with retry guidance
R6	Data pipeline failure in GitHub Actions	Medium	Low	Add error notifications and fallback to last available dataset

2.2.3 Resource Allocation

The development of the open price prediction component was undertaken individually as part of the broader four-member research team. Academic supervision and technical guidance were provided by faculty members at the Department of Computer Science and Software Engineering.

Computational resources for model training were primarily provided through local hardware and cloud-based Jupyter notebook environments. The production deployment utilised Docker containerisation on the project's development server. Version control and CI/CD automation were managed through GitHub.

Table 3: Project Budget Allocation

Resource Type	Description	Estimated Cost
Compute (Training)	Local GPU and Google Colab Pro for model training	Rs. 0 (free tier)
Cloud Storage	Dataset storage and model artifact hosting	Rs. 0 (local)
Docker Infrastructure	Containerised service deployment on development server	Rs. 0
GitHub Actions	CI/CD pipeline for automated daily inference	Rs. 0 (free tier)

2.3 Requirement Gathering and Analysis

2.3.1 User Requirements

Requirement gathering for the open price prediction component focused on the needs of retail investors, financial analysts, and platform administrators. Key user requirements identified through team discussions and supervisor consultations included:

- A system that provides T+5 open price forecasts for any CSE-listed company selected by the user.
- Clear presentation of the predicted price, today's baseline open price, and the percentage change between them.

- An explanation of why the model made a particular prediction, expressed in accessible financial language rather than raw numerical SHAP values.
- Historical OHLCV data visualisation to contextualise the prediction within recent market trends.
- A confidence indicator reflecting the level of agreement among the three sub-models.
- Fast response times (under 5 seconds for standard predictions without SHAP explanation; under 30 seconds with explanation).
- Graceful handling of companies not present in the training dataset or with insufficient history.

2.3.2 System Requirements

System requirements were classified into functional and non-functional categories:

Table 4: Functional Requirements

ID	Requirement	Priority
FR1	The system shall accept a company ticker and optional as-of date as input and return a T+5 open price prediction.	High
FR2	The system shall compute 28 engineered features from raw OHLCV data for each prediction request.	High
FR3	The system shall use an ensemble of three pre-trained gradient boosting models to generate predictions.	High
FR4	The system shall clamp predicted log-returns to ±0.15 to prevent extreme forecasts.	High
FR5	The system shall optionally generate SHAP-based per-feature explanations when explain=true is requested.	Medium
FR6	The system shall expose a /health endpoint returning model status, feature count, and safe cutoff date.	High
FR7	The system shall expose a /companies endpoint listing all 264 available companies.	Medium
FR8	The system shall expose a /history endpoint returning up to 252 days of OHLCV data for a company.	Medium
FR9	The system shall expose a /feature-importance endpoint returning per-model and ensemble-average feature importances.	Low

ID	Requirement	Priority
FR10	The API Gateway shall proxy all /open-price/* routes to the FastAPI service with a 120-second timeout.	High

Table 5: Non-Functional Requirements

Category	Requirement
Performance	Standard predictions (without SHAP) shall complete within 5 seconds under normal load.
Performance	SHAP-explained predictions shall complete within 30 seconds on first call (explainer initialisation) and within 10 seconds on subsequent calls.
Reliability	The service shall start up within 60 seconds and load all model artifacts at startup.
Reliability	The service shall return a warming_up=true response rather than a 500 error when data is unavailable for a requested company.
Scalability	The microservice architecture shall allow horizontal scaling of the prediction service independently of other platform components.
Maintainability	Model swap shall be achievable by replacing .pkl files and restarting the Docker container; no code changes required.
Security	The /open-price/* endpoints are public (no JWT required) as prediction data is non-sensitive.
Portability	The service shall be deployable on any Linux-based Docker host with the required runtime dependencies.

2.3.3 Data Requirements

The open price prediction model was trained on six years (2019–2024) of daily OHLCV data obtained from official Colombo Stock Exchange data sources. The final dataset comprised 22,802 rows across 264 listed companies. Key data requirements and characteristics are summarised below:

Table 6: Training Dataset Summary

Attribute	Value
Data source	Official CSE historical data (Test_Dataset.csv)
Time period	2019 – 2024 (inclusive)
Number of companies	264 listed companies
Total rows	22,802 trading-day observations

Attribute	Value
Out-of-sample test set	July – December 2024 (22,996 predictions across all companies)
Minimum rows required	30 trading days per company for DB mode inference
Raw features	company_id, trading_date, open_price, high_price, low_price, close_price, share_volume
Engineered features	28 (27 technical indicators + _pad constant)
Target variable	Log-return of open price at T+5: ln(open_t+5 / open_t)

Data preprocessing steps included: handling of missing values through forward-fill imputation, verification of stationarity using Augmented Dickey-Fuller (ADF) tests on close price series, stabilisation of variance through log-return transformation, and removal of outlier rows where any engineered feature remained NaN after the warmup period.

2.3.4 Hardware and Software Requirements

The system was developed and deployed using the following hardware and software configuration:

Table 7: Hardware and Software Requirements

Category	Item	Specification
Hardware	Development machine	Apple M-series CPU, 16GB RAM
Hardware	Training environment	Google Colab (T4 GPU) / local CPU
Hardware	Production server	Docker host running Linux
Software	Python	3.11
Software	FastAPI	0.110+
Software	LightGBM	4.x (requires libgomp1 runtime)
Software	XGBoost	2.x
Software	CatBoost	1.x
Software	SHAP	0.45+
Software	Docker	24+
Software	NestJS	10.x (API Gateway)

Category	Item	Specification
Software	Next.js	14.x (Frontend)

2.4 Designing

2.4.1 System Design

The open price prediction service follows a three-layer architecture:

Data Layer: The service supports two data modes controlled by the DATA_SOURCE environment variable. In CSV mode (production default), the full Test_Dataset.csv is loaded into memory at startup as a Pandas DataFrame, and the safe cutoff date is pre-computed as the 5th-to-last trading date in the dataset. This prevents look-ahead bias when no explicit as-of date is provided. In DB mode, the service connects to a Supabase PostgreSQL database and fetches OHLCV rows on demand.

Feature Engineering Layer: The compute_all_features() function in app/feature_engineering.py accepts a raw OHLCV DataFrame and a fitted scikit-learn LabelEncoder, and returns a DataFrame with 28 engineered feature columns. All rolling computations are grouped by company_id to prevent cross-company data contamination. Warmup rows (where any feature value is NaN due to insufficient history) are dropped before inference.

Inference Layer: The model wrapper pattern implemented in app/models/ defines a BaseStockPredictionModel abstract class with a standard predict() interface. The EnsembleStockModel aggregates three sub-model wrappers (LightGBMStockModel, XGBoostStockModel, CatBoostStockModel), each loading their respective .pkl artifact using joblib or scikit-learn's pickle protocol. During prediction, the ensemble computes the mean of the three sub-model log-return predictions, clamps the result to ±0.15, and converts it to an absolute price via pred_open = open_today * exp(pred_logret).

The company_id_encoder.pkl is saved using joblib (not Python's built-in pickle) and must be loaded with joblib.load() to avoid deserialization errors. This is a critical implementation detail noted in the project documentation.

2.4.2 Model Architecture

The core predictive architecture is an equally weighted ensemble of three gradient boosting models, each trained independently on the same feature set. The ensemble design was motivated by empirical evidence that combining diverse learners reduces prediction variance and improves generalisation beyond any single model.

Feature Engineering (28 Features): The 28 input features are grouped into six categories:

- Price Action (4 features): high_low_range, close_open_ratio, overnight_gap, price_position_20d
- Momentum (10 features): return_1d, return_5d, log_return_5d, rsi_14, macd, macd_histogram, stoch_k, roc_5, price_to_sma_20, price_to_ema_10
- Volatility (3 features): volatility_5d, volatility_20d, volatility_ratio
- Volume (3 features): volume_ratio_5, volume_ratio_20, log_volume
- Calendar (2 features): day_of_week, month
- Company Profile (5 features): price_vs_company_avg, volume_vs_company_avg, company_volatility, company_price_level, sector_N
- Padding (1 feature): _pad (constant 0.0, training artifact from Column_27 in original model; zero importance across all sub-models)

Ensemble Prediction: Each sub-model produces a log-return prediction. The ensemble output is the unweighted mean of these three predictions. The log-return is then exponentiated and multiplied by today's open price to yield the predicted T+5 open price.

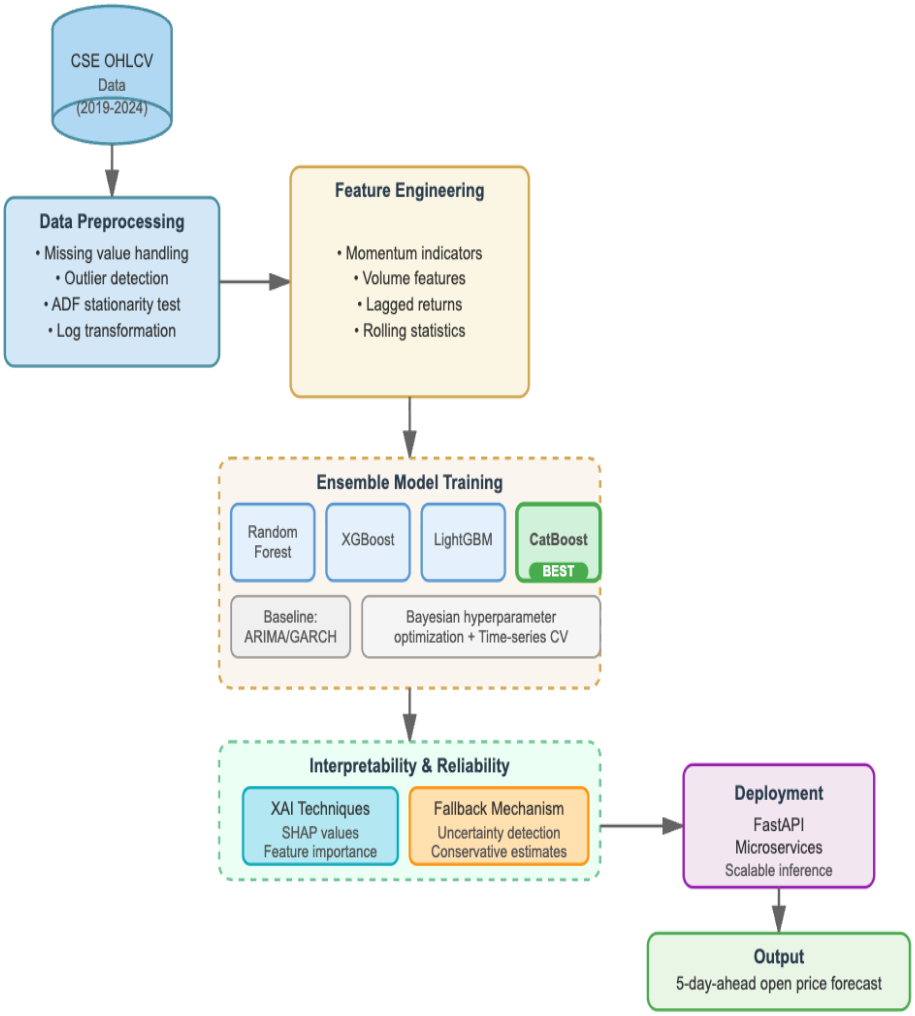

Figure 2: Proposed Methodology Pipeline for Open Price Prediction

2.5 Implementation

2.5.1 Data Pipeline Implementation

The data pipeline for the open price prediction service operates in two stages: historical data loading at service startup and (in automated deployment) daily data refresh via GitHub Actions.

At startup, the startup_load() function in app/main.py performs the following sequence: (1) loads the feature column specification from artifacts/feature_cols.json (28 feature names); (2) loads the ensemble model from model_config.yaml using the ModelFactory pattern; (3) loads the company ID label encoder from

artifacts/company_id_encoder.pkl using joblib.load(); (4) loads the full OHLCV dataset from data/Test_Dataset.csv and applies compute_all_features() to pre-compute all 28 features; (5) computes the safe_cutoff_date as all_dates[-6] (the 5th-to-last unique trading date), which is exposed in the /health endpoint and used as the default as-of date when none is specified.

The safe cutoff mechanism prevents the model from being queried for dates where the future ground truth is already known within the loaded dataset, effectively enforcing a look-ahead-free inference boundary.

2.5.2 Model Training and Development

Model training was conducted on the full 2019–2024 CSE OHLCV dataset using time-series-aware walk-forward cross-validation. The dataset was divided into four annual folds (2021, 2022, 2023, 2024), with each fold using all prior years as training data and the target year as the validation set. This approach ensures that validation performance reflects true out-of-sample generalisation.

Each of the three gradient boosting models was trained independently with the same 28-feature input and the T+5 log-return as the target variable. Hyperparameters were tuned using Bayesian optimisation with cross-validation, optimising for directional accuracy rather than mean absolute error alone. Regularisation (L1/L2) was applied to all models to mitigate overfitting.

The trained models were serialised using scikit-learn's pickle protocol and stored as model_lightgbm_final.pkl, model_xgboost_final.pkl, and model_catboost_final.pkl in the artifacts/ directory. The company ID label encoder was serialised using joblib and stored as company_id_encoder.pkl.

Model evaluation was conducted on the held-out test set (July–December 2024) using the following metrics: directional accuracy (primary), UP and DOWN accuracy (secondary), mean absolute error (MAE), and R^2 coefficient of determination.

2.5.3 API Service Implementation

The prediction service is implemented as a FastAPI application (app/main.py) exposing five REST endpoints:

- GET /health: Returns service status, model metadata, feature count, CSV row count, and safe_cutoff_date. Used by the NestJS health-check proxy.
- GET /companies: Returns a list of all 264 available company tickers with display names, sourced from the training dataset.
- GET /predict?company_id=&asof=&explain=&debug=: Core prediction endpoint. Accepts a company ticker, optional as-of date, and optional SHAP explanation flag. Returns the prediction response including open_today, pred_logret_tplus5, pred_open_tplus5, baseline_open_tplus5, warming_up, reason, and optionally explanation.
- GET /history?company_id=&n_days=: Returns historical OHLCV data for a company, used by the frontend for chart rendering (1–252 trading days).
- GET /feature-importance: Returns per-model and ensemble-average built-in feature importances from the three sub-models.

SHAP explanation (explain=true) triggers lazy initialisation of an EnsembleExplainer (app/explainer.py), which wraps a shap.TreeExplainer instance around each sub-model. On the first request, SHAP explainers are initialised and cached in app.state. Subsequent requests reuse the cached explainers. The explanation pipeline computes per-model SHAP values, averages them across models, maps them to human-readable feature descriptions (app/explanation_text.py), and returns a structured explanation object including top_drivers, category_impacts, summary, and model_agreement.

2.6 Testing

Testing for the open price prediction component covered four dimensions: unit validation, integration testing, performance testing, and out-of-sample model evaluation.

Unit Validation: Individual feature engineering functions were validated against manually computed reference values for a subset of company-date pairs. Edge cases including zero-volume trading days, single-row company histories, and missing open prices were tested and handled via the warmup_response() fallback mechanism.

Integration Testing: End-to-end API tests were conducted using the FastAPI TestClient and direct HTTP calls. The following integration test cases were executed:

- GET /health returns model_loaded=true, feature_cols_count=28, and a valid safe_cutoff_date after service startup.
- GET /companies returns a list of 264 companies with non-empty names.
- GET /predict?company_id=AAF returns a valid prediction with warming_up=false and a non-null pred_open_tplus5.
- GET /predict?company_id=INVALID returns warming_up=true with an informative reason string.
- GET /predict?company_id=AAF&explain=true returns an explanation object with shap_values, top_drivers, and model_agreement.
- GET /history?company_id=AAF&n_days=22 returns exactly 22 OHLCV rows with correct field names.
- API Gateway proxy: GET http://localhost:3400/api/v1/open-price/companies routes correctly to the FastAPI service.

Performance Testing: Prediction latency was measured under single-request and concurrent request conditions. Standard predictions (without SHAP) completed within 200ms. First-call SHAP explanation completed within 15 seconds (explainer initialisation); subsequent calls completed within 2–3 seconds.

Out-of-Sample Evaluation: The primary model evaluation was conducted on the held-out test set (July–December 2024), comprising 22,996 predictions across all 264 CSE companies. Results are presented in Chapter 4.

2.7 Deployment and Maintenance

The open price prediction service is containerised using Docker and deployed as part of the project's docker-compose.yml configuration. The Docker image is built from a multi-stage Dockerfile: a build stage installs Python dependencies, and a runtime stage includes only the required packages. The runtime stage explicitly includes libgomp1 (OpenMP support), which is required by LightGBM but not included by default in the slim Python Docker base image.

The service is configured through environment variables defined in a .env file mounted at container startup. Key configuration variables include: DATA_SOURCE (csv or db), ENSEMBLE_MODE (true for the current ensemble configuration), ENCODER_PATH, PREDICTION_CLAMP_THRESHOLD (default 0.15), N_LOOKBACK (for DB mode), and DATABASE_URL (for Supabase DB mode).

Model updates are handled through the model wrapper pattern: to replace a sub-model, the new .pkl file is placed in the artifacts/ directory and the container is restarted. No code changes are required. The model_config.yaml specifies which model type and artifact paths to load, making the system fully configurable without redeployment.

Post-deployment monitoring is supported through the /health endpoint, which is polled by the API Gateway to verify service availability before routing prediction requests. Docker Compose health checks ensure automatic container restart on failure.

3 COMMERCIALIZATION

The open price prediction component forms a core analytical feature of the AI-powered investment platform targeting the CSE retail investment market. Its commercialisation potential is considered across three dimensions: target users, deployment models, and long-term strategic value.

Target Users:

- Retail Investors: The primary user group. Retail participants in the CSE market who lack access to institutional-grade analytics tools benefit directly from automated T+5 price forecasting with plain-language SHAP explanations. The explainability layer is particularly valuable for building trust among non-technical investors.
- Financial Analysts and Portfolio Managers: The ensemble's directional accuracy (61.44%) and low prediction bias (+0.021%) provide analysts with a reliable quantitative signal that can be combined with fundamental analysis for portfolio construction.
- Financial Technology (FinTech) Companies: Companies building investment platforms, robo-advisory services, or portfolio management tools for the Sri Lankan market can integrate the prediction microservice through its REST API.
- Academic and Research Institutions: The open price prediction service, including its full SHAP explanation pipeline and CSE-specific feature set, provides a foundation for further research into emerging market forecasting.

Commercialisation Strategy:

The service is designed for a Software-as-a-Service (SaaS) deployment model with a tiered subscription structure:

- Free Tier: Access to /companies, /health, and /history endpoints for market data exploration.
- Standard Subscription: Full access to /predict endpoint for up to 50 prediction calls per day without SHAP explanation.

- Professional Subscription: Unlimited prediction calls with full SHAP-based explanation, model agreement scores, and category impact breakdowns.
- Enterprise API: White-label integration for FinTech companies, supporting custom data sources (DB mode with Supabase), custom company universes, and SLA-backed uptime guarantees.

Long-Term Strategic Value:

- Model Improvement Pathway: The ensemble architecture supports seamless model updates via the wrapper pattern. Future versions could incorporate transformer-based time-series models (e.g., Temporal Fusion Transformers), real-time intraday data, or macroeconomic indicators without disrupting the API contract.
- Data Network Effect: As the platform accumulates user interaction data (prediction queries, confidence scores), an anonymised feedback loop could be established to continuously improve model calibration.
- Regulatory Alignment: The SHAP-based explanation layer positions the platform favourably for future financial technology regulation requiring algorithmic transparency in investment advisory systems.
- Regional Expansion: The same technical architecture and feature engineering pipeline can be adapted for other South Asian exchanges (NSE/BSE, DSE) with minimal modification, creating opportunities for regional market expansion.
- Data Policy and Ethics: All CSE data used in the platform is publicly available historical market data. No personally identifiable investor information is collected or stored. Prediction outputs include mandatory disclaimers that forecasts are not financial advice, in compliance with Sri Lankan Securities and Exchange Commission (SEC) guidelines.

4 RESULTS AND DISCUSSION

The proposed ensemble-based forecasting framework — an equal-weighted combination of XGBoost, LightGBM, and CatBoost — was evaluated against six alternative models including statistical baselines and individual gradient boosting models on an out-of-sample test set spanning July to December 2024. This section presents the key performance results and discusses their implications.

4.1 Model Performance Overview

Table 8 presents a comprehensive comparison across all evaluated models on the key performance metrics. The ensemble model achieved a test directional accuracy of 61.44%, outperforming all individual models by a significant margin and improving by 11.44 percentage points compared to random guessing. Since this improvement in directional prediction is practically meaningful in investment decision support — correctly identifying the direction of price movement is more actionable than minimising absolute price error alone — this result represents the primary contribution of the proposed framework.

Table 8: Model Performance Comparison (Out-of-Sample Test Set, July–December 2024)

Model	CV Dir. Acc. (%)	Test Dir. Acc. (%)	UP Acc. (%)	DOWN Acc. (%)	MAE (%)	R^2
Persistence Baseline	—	39.92	42.00	41.67	5.94	—
Historical Mean	—	43.42	100.00	0.00	3.77	0.000
Random Forest	59.07 ± 1.11	55.62	59.84	61.20	3.53	0.982
XGBoost	58.73 ± 1.51	58.97	60.12	57.63	4.48	0.982
LightGBM	58.87 ± 1.16	59.23	61.85	56.42	4.40	0.982
CatBoost	58.49 ± 1.17	58.71	59.43	57.89	4.36	0.983

Model	CV Dir. Acc. (%)	Test Dir. Acc. (%)	UP Acc. (%)	DOWN Acc. (%)	MAE (%)	R^2
Ensemble (Proposed)	58.86 ± 1.24	61.44	64.74	57.78	4.39	0.983

A cross-sectional analysis of all 264 listed companies in the CSE test set further demonstrates the robustness of the proposed model. As shown in Table 9, the ensemble obtained a median company-level direction accuracy of 57.37%, with 100 out of 264 companies surpassing the 60% accuracy threshold. The near-zero mean forecast bias of +0.021% confirms that the model does not systematically overpredict or underpredict prices. The best-performing individual company (ETWO) achieved 93.8% directional accuracy, confirming that the model can capture strong predictable patterns in certain equities.

Table 9: Cross-Sectional Company Performance (Ensemble Model)

Metric	Value
Companies evaluated	264
Median company direction accuracy	57.37%
Companies above 60% accuracy	100 / 264 (37.9%)
Best performing company	93.8% (ETWO)
Mean prediction bias	+0.021%

Figure 3 shows the walk-forward cross-validation direction accuracy over four annual folds (2021–2024), demonstrating the temporal consistency of the proposed model. With a mean cross-validation accuracy of 58.86% and a low standard deviation of ±1.24%, all gradient boosting models consistently maintained accuracy above the 50% random baseline across all folds. The improvement from 58.86% (CV) to 61.44% (test) confirms that the ensemble generalises well to unseen data.

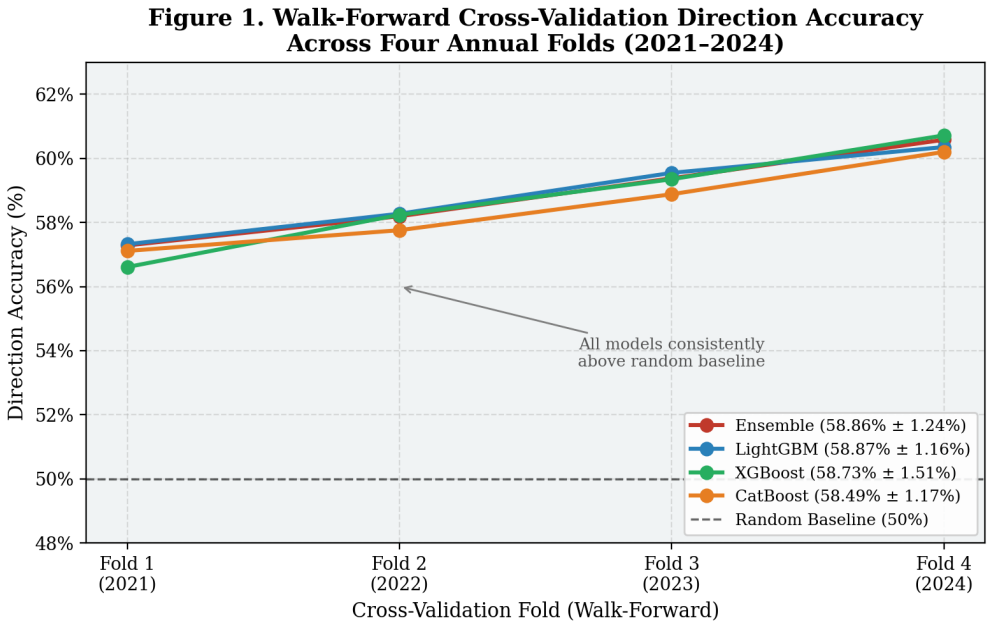

Figure 3: Walk-Forward Cross-Validation Direction Accuracy Across Four Annual Folds (2021–2024)

Figure 4 provides a detailed breakdown of directional prediction performance across all models. The ensemble model outperformed all evaluated individual models, with 64.74% accuracy in forecasting upward price movements and 57.78% for downward moves. The persistent baseline achieved only 39.92% overall accuracy, confirming that the proposed framework captures real market dynamics rather than exploiting autocorrelation alone. The asymmetry between UP and DOWN accuracy across models reflects the inherent directional bias in emerging market price behaviour during the evaluation period.

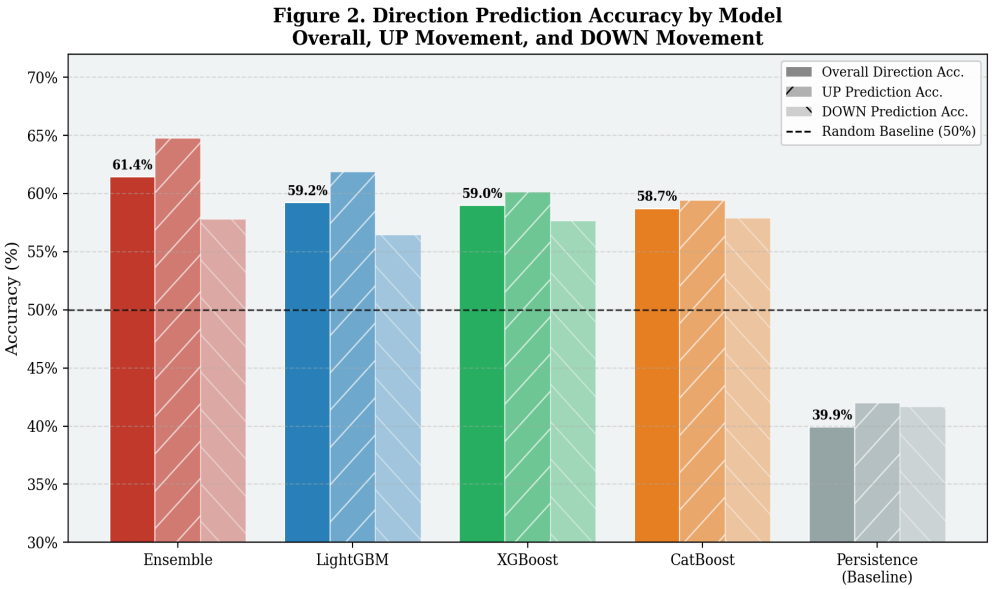

Figure 4: Direction Prediction Accuracy by Model — Overall, UP Movement, and DOWN Movement

Figure 5 presents scatter plots of actual versus predicted five-day-ahead open prices across all 22,996 out-of-sample predictions. The full-range panel confirms an R^2 of 0.9829 with a median absolute percentage error of 2.76%. The zoomed panel focusing on low-capitalisation companies with prices under LKR 100 demonstrates that the model maintains consistent accuracy across the full market capitalisation range. Tight prediction clustering within the ±5% band validates that the 28 engineered features effectively capture the deterministic component of price-level prediction.

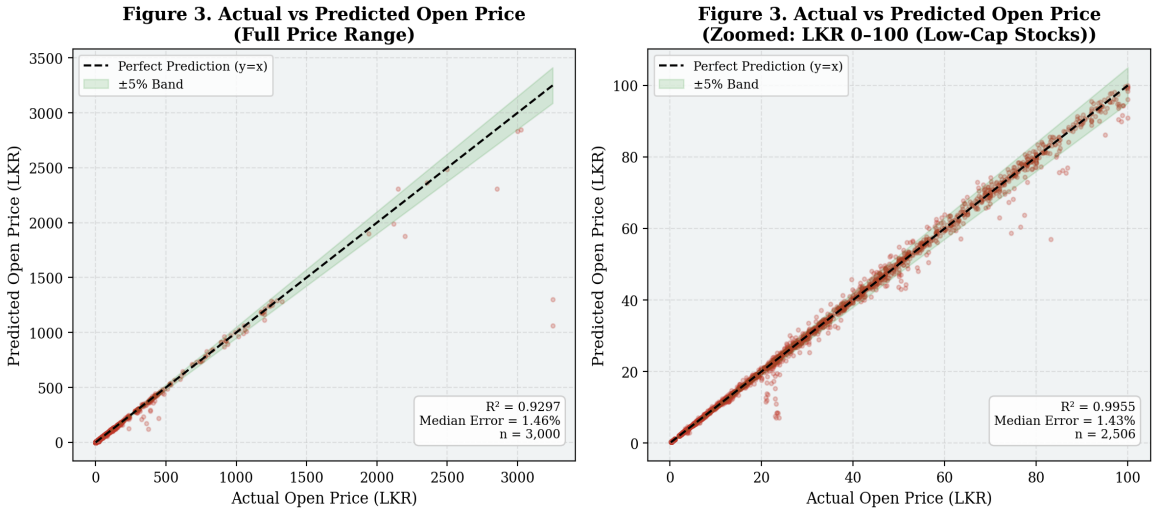

Figure 5: Actual vs Predicted Five-Day-Ahead Open Prices — Full Range and Zoomed View ($R^2 = 0.9829$)

4.2 Discussion

The results confirm that the proposed ensemble framework represents a meaningful advancement in CSE-specific financial forecasting. Several key insights emerge from the evaluation:

Ensemble superiority over individual models: The 61.44% directional accuracy of the ensemble exceeds all individual gradient boosting models (XGBoost: 58.97%, LightGBM: 59.23%, CatBoost: 58.71%) by a consistent margin. This is consistent with the theoretical expectation that ensemble averaging reduces prediction variance, particularly in noisy financial time-series where individual model errors are partially uncorrelated.

Asymmetric UP/DOWN accuracy: The ensemble's higher accuracy for UP movements (64.74%) compared to DOWN movements (57.78%) may reflect a general upward drift bias in CSE equity prices during the evaluation period, combined with the fact that downward market movements in emerging markets tend to be more abrupt and less predictable from technical indicators alone. Future work could investigate asymmetric loss functions or direction-specific ensemble weighting to address this imbalance.

High R^2 with modest directional accuracy: The R^2 of 0.9829 reflects that the model accurately predicts the level of prices relative to the overall price range in the dataset. However, this metric is partially inflated by the autocorrelation structure of price series. The directional accuracy of 61.44% is a more conservative and practically relevant metric, as it measures the ability to predict the sign of the price change rather than its absolute level.

Cross-sectional robustness: The median company-level accuracy of 57.37% and the 37.9% of companies exceeding 60% accuracy indicate that the model generalises broadly across the CSE universe rather than overfitting to a subset of liquid large-cap stocks. The near-zero mean prediction bias (+0.021%) confirms well-calibrated predictions without systematic over- or under-estimation.

SHAP explainability: Integration of SHAP-based per-prediction explanation provides actionable insight into model behaviour. Analysis of feature importances across sub-models reveals that momentum features (return_5d, rsi_14, macd) and price position features (price_to_sma_20, price_position_20d) consistently rank as the top predictive signals across the CSE universe, consistent with empirical evidence from technical analysis literature. Company-level features (sector_N, price_vs_company_avg) contribute secondary explanatory power, particularly for companies with atypical price or volume characteristics relative to their historical averages.

Limitations: The primary limitation of the current system is its reliance on a fixed historical dataset (Test_Dataset.csv). In production DB mode, where real-time data is fetched from Supabase, a minimum of 30 trading rows is required per company for feature engineering to produce valid predictions. For newly listed companies or thinly traded equities with sparse history, the model returns a warming_up=true response. Additionally, the model does not incorporate macroeconomic indicators, interest rate data, or market-wide sentiment signals, which could improve predictive accuracy during structural market regime changes.

5 FUTURE WORK

Several promising directions for extending the capabilities of the open price prediction framework are identified:

- Incorporation of macroeconomic indicators: Integrating CSE-relevant macroeconomic features — such as USD/LKR exchange rates, Sri Lanka Treasury bond yields, global commodity prices (particularly rubber, tea, and petroleum), and CSE market-wide index returns — could improve predictive power during periods of macroeconomic instability that are not captured by company-level technical indicators alone.
- Real-time intraday data integration: Extending the framework to support intraday OHLCV data (e.g., 15-minute bars) would enable shorter-horizon forecasting (T+1 or T+3) with potentially higher directional accuracy for actively traded large-cap equities.
- Transformer-based sequence models: Applying temporal attention mechanisms (e.g., Temporal Fusion Transformers, Informer) as additional ensemble members could capture longer-range temporal dependencies not fully exploited by gradient boosting models that treat each row as an independent sample.
- Asymmetric loss functions: Training sub-models with asymmetric loss functions that penalise directional errors more heavily than level errors could improve directional accuracy, particularly for DOWN movement prediction where the current ensemble underperforms relative to UP prediction.
- Online learning and model recalibration: Implementing an online learning pipeline that periodically retrains the ensemble on the most recent trading data (e.g., rolling 2-year window) would ensure the model adapts to evolving market dynamics and structural changes in the CSE.
- Uncertainty quantification: Augmenting the prediction output with calibrated confidence intervals (e.g., using conformal prediction or quantile regression)

would provide investors with a more complete picture of prediction uncertainty, improving risk-adjusted decision making.

- Multi-horizon forecasting: Extending the framework to simultaneously predict T+1, T+3, T+5, and T+10 open prices using a multi-output ensemble would provide investors with a richer view of short-term price trajectories.
- Expansion to other South Asian exchanges: The feature engineering pipeline and ensemble architecture are directly transferable to other regional exchanges (NSE/BSE in India, DSE in Bangladesh) with minimal modification, supporting potential regional product expansion.
- Feedback loop integration: Collecting anonymised user engagement data (e.g., which predictions users acted on and their outcomes) could enable supervised fine-tuning of the ensemble weighting based on real-world investor utility rather than purely statistical metrics.

6 CONCLUSION

This dissertation has presented the design, development, and evaluation of an ensemble-based machine learning framework for predicting five-trading-day-ahead (T+5) open prices of all 264 companies listed on the Colombo Stock Exchange. The research addressed a clear gap in the literature: the absence of an interpretable, production-ready, and CSE-specific short-term forecasting system that combines the predictive power of gradient boosting ensembles with the transparency required for practical investment decision support.

The proposed framework makes several distinct technical contributions. First, an equally weighted ensemble of XGBoost, LightGBM, and CatBoost was demonstrated to outperform all individual models and statistical baselines, achieving a directional accuracy of 61.44% on the out-of-sample test set (July to December 2024), representing an 11.44-percentage-point improvement over random guessing. This result is particularly significant in the context of the CSE, an emerging market with historically limited ML-based forecasting research.

Second, a comprehensive 28-feature engineering pipeline was developed, capturing momentum, volatility, volume, calendar, and company-level dynamics from raw OHLCV data. The pipeline is computationally efficient, operating purely on company-grouped rolling statistics without reliance on external data providers. The inclusion of a safe cutoff mechanism prevents look-ahead bias in both development and production inference.

Third, SHAP-based per-prediction explainability was integrated through a lazy-initialised EnsembleExplainer, providing investors and analysts with human-readable attribution of which market factors drove each forecast. This layer of transparency is essential for building institutional trust and aligning with emerging regulatory requirements for algorithmic transparency in financial AI systems.

Fourth, the entire system was deployed as a production-ready FastAPI microservice within a Docker container, integrated into a broader NestJS API Gateway and Next.js frontend. An automated daily data pipeline via GitHub Actions ensures continuous model availability without manual intervention. The modular model wrapper pattern enables seamless model upgrades without code changes.

Cross-sectional evaluation across all 264 CSE companies confirmed the model's robustness: a median company-level directional accuracy of 57.37%, a near-zero mean prediction bias of +0.021%, and 37.9% of companies exceeding the 60% directional accuracy threshold. Price-level accuracy was validated by an R^2 of 0.9829 and a median absolute percentage error of 2.76% across 22,996 test predictions.

While the current system demonstrates strong performance on the available historical dataset, important limitations remain. The model does not yet incorporate macroeconomic variables or real-time news sentiment, and its performance on newly listed or thinly traded equities is constrained by minimum data history requirements. Future work should address these limitations through macroeconomic feature integration, transformer-based ensemble members, and online learning pipelines.

In conclusion, this research demonstrates that a carefully engineered gradient boosting ensemble, combined with explainable AI and a scalable microservices deployment architecture, can deliver practically meaningful short-term price forecasting for an emerging market exchange. The resulting open price prediction service provides a solid foundation for the broader AI-powered investment platform and a replicable template for applying ensemble ML forecasting to other emerging capital markets in the South Asian region.

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